

Candidate Name: _____ Class: 07S_____



JURONG JUNIOR COLLEGE

PRELIMINARY EXAMINATION 2008

H1 BIOLOGY 8875/2

FRIDAY

15 AUGUST 2008

2 HOURS

INSTRUCTIONS TO CANDIDATES :

Write your name and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working. Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answer in the spaces provided on the question paper.

Section B

Answer any **one** question.

At the end of the examination

1. Fasten any separate answer paper used securely to the question paper.
2. Enter the question number from Section B in the grid opposite.

EXAMINER'S USE ONLY	
Section A	Mark
1	
2	
3	
4	
Section B	
Total	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer **all** the questions in this section.

1. During digestion, large molecules are broken down into their constituent monomers (subunits).

(a) Complete the table below. [2]

Large molecules	Monomers (subunits)
Protein ; <i>Accept named eg.</i>	Amino acids
Lipid	Fatty acids and glycerol ;
Polysaccharide ; <i>Reject: 'carbohydrates'</i> <i>Accept named eg.</i>	Monosaccharide
DNA	Deoxyribonucleotides ;

Students were awarded two marks for four correct answers; minus one mark for every mistake made.

(b) Explain how a monosaccharide enters an epithelial cell. [2]

A monosaccharide moves by facilitated diffusion or active transport;;

• This occurs via channel proteins or by carrier proteins molecules;;

Reject "endocytosis" → bulk transport

Starch, a storage molecule in plants, is a polymer made of glucose molecules joined together. It consists of two components.

(c) Name these two components. [2]

- amylose;;**
- amylopectin;;**

(d) Give one similarity and one difference between these two components. [2]

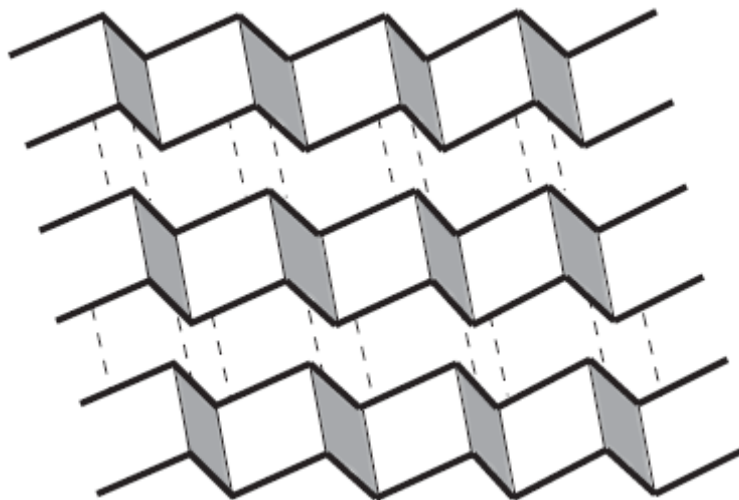
Similarity (Either 1)

- contain α -glucose monomers;;**
- coiled into a helix;;**
- no cross-linking;;**
- contain $\alpha(1\rightarrow4)$ glycosidic bonds;;**

Difference (Either 1)

- The amylose component is unbranched whilst the amylopectin component is branched;;**
- The glucose residues in the amylose component are linked by $\alpha(1\rightarrow4)$ glycosidic bonds, while the glucose residues in the amylopectin component are linked by both $\alpha(1\rightarrow4)$ glycosidic bonds and $\alpha(1\rightarrow6)$ glycosidic bonds;;**

Protein molecules come in many shapes and forms that can be classified into primary, secondary, tertiary and quaternary. The secondary, tertiary and quaternary shapes arise as a result of different kinds of folding of a primary structure. One kind of secondary structure is a pleated sheet where the primary molecule extends along the folded sheet. The primary structures in the layers are held together by hydrogen bonding.



(e) Suggest two ways in which such a structure may be important to the functioning of the protein. [2]

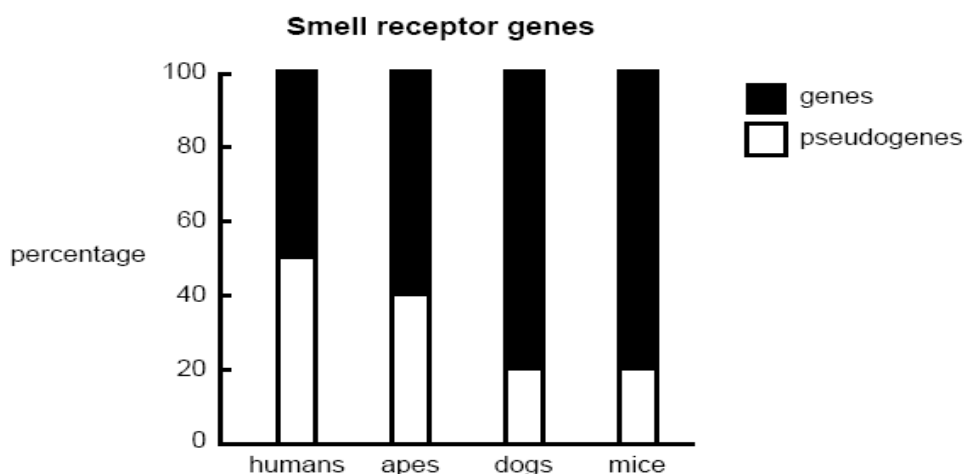
Structures may be important because:

- of the ability of the protein to stretch or contract / elongate or shorten / be flexible in particular situations;;
- pleating may strengthen the molecule / provide tensile strength and that may be important for its function;;
- particular structures may provide an active or binding site for an enzyme or other molecule;;

[Total: 10]

2. Pseudogenes are the remains of broken genes which are unable to function and can be considered to be genetic fossils. Some are relics of genes lost through evolution while others reflect an earlier version of a present functional gene. Pseudogenes are able to accumulate all kinds of mutations.

In mammals, more than one thousand genes coding for smell receptors have been identified. Individual species vary in the proportion of these genes that have become pseudogenes, and humans have fewer than 500 functional genes coding for smell receptors. The majority of these genes are still functional genes in rats and mice.



- (a)
(i) From the **graph**, what can be **concluded** about the importance of smell to the survival of these **four groups**? [2]

- Smell is **most** important to dogs and mice;;
- **less** important to apes and **least** important to humans;;

NOTE: Students are required to use the names of the specific examples given rather than making general statements.

- (ii) Mice and dogs can only distinguish shades of grey, while some apes and monkeys are able to distinguish colours. **Would you expect to find any change** in the numbers of smell pseudogenes in apes and monkeys able to distinguish colours? Explain why. [2]

- **Yes;;**
- **Apes and monkeys are less dependent on smell / more dependent on vision for survival;**
- **They would therefore have more pseudogenes / less smell genes;**

NOTE: Students are to make the link that, as smell is less important to these animals, mutations are possible in genes responsible for smell as these will not affect the organism's chance of survival. These mutated genes are classed as pseudogenes.

Senses/systems work together to ensure survival of species, even though genes involved may be located at separate gene loci!

- (iii) Why are pseudogenes able to accumulate mutations that do not exist in functional genes? [2]

- **Pseudogenes are not expressed / non-coding and any mutation in them does not harm the offspring when passed on;;**
- **Functional genes become disrupted / produce a different protein which could affect the survival of offspring, thus less likely to be passed on (idea to be conveyed at least once) to subsequent generations;;**

Body cells have elaborate control systems that regulate their division. If these control systems break down, cells start to grow and divide in an unregulated way and produce a tumour.

- (b) Cigarette smoking can lead to the formation of tumours. Describe how. [4]

- **Contains carcinogens/cancer-causing substances;**
- **Causes mutation/changes in DNA/genetic material;**
- **Of genes controlling normal rate of cell division/regulating cell cycle;**
- **Tumour suppressor genes; becomes less active / loss-of-function;**
- **Proto-oncogenes; become oncogenes / more active / gain-of-function; (if only 'tumour suppressor genes' and 'proto-oncogenes' stated without mention of activity levels → 0.5 marks)**
- **Leads to uncontrolled cell division/mitosis;**

[Total: 10]

3. Cytochrome c is found in the mitochondria of every aerobic eukaryote — animal, plant, and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. **These molecules are homologous.**

Table 8 shows the first 18 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath. A ditto (“) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

Cytochrome	Amino acid																		
		1					6				10				14			17	18
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His
Pig		""	""	""	""	""	""	""	""	""	""	Val	Gln	""	""	Ala	""	""	""
Chicken		""	""	Ile	""	""	""	""	""	""	""	Val	Gln	""	""	""	""	""	""
Dogfish		""	""	""	""	""	""	""	""	Val	""	Val	Gln	""	""	Ala	""	""	""
<i>Drosophila</i>	<<<	""	""	""	""	""	""	""	""	Leu	""	Val	Gln	Arg	""	Ala	""	""	""
Wheat	<<<	""	Asn	Pro	Asp	Ala	""	Ala	""	""	""	Lys	Thr	""	""	Ala	""	""	""
Yeast	<<<	""	Ser	Ala	Lys	""	""	Ala	Thr	Leu	""	Lys	Thr	Arg	""	""	""	""	""

(a) With reference to the table above, explain what is meant by “These molecules are homologous”. [3]

- The cytochrome molecule of the various organisms is a shared characteristic derived from a common ancestor ;;
- The cytochrome molecules share the same amino acids at many positions ;;
- For all organisms, the molecules have the same amino acid at positions 1,6,10,14, 16, 17 and 18, from the N-terminal ;;

(b) (i) Explain why the analysis of amino acid sequence is used to establish evolutionary relationship. [3]

- The amino acid sequence of a protein is defined by the DNA sequence;;
- Mutations to the cytochrome c gene became fixed in a population / part of the genome and were inherited ;;
- Over time, as each population accumulates its own set of mutations, the original DNA sequence diverges and is reflected in the amino acid sequence. ;;
- The more similar the amino acid sequence, the more recently the divergence took place / more closely related the two organisms are ;;

[max 3]

(ii) State a limitation of using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

- One protein represents only a very small percentage of the genetic information in the organisms;;
- Is not representative of the entire genome;;
- Data from one protein may not be accurate;;

Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for humans and the apes (e.g. chimpanzee), their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(c) State the name given to

Coding regions **exons**;;

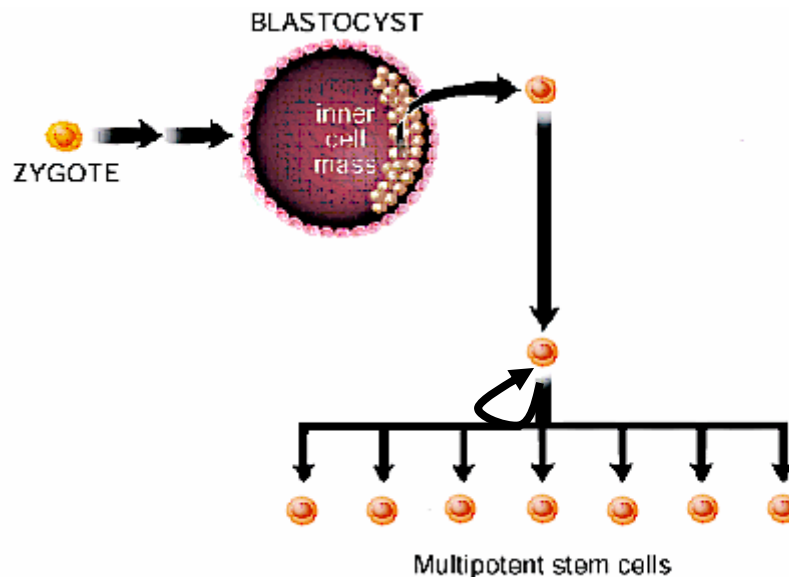
Non-coding regions **introns**;; [2]

(d) Suggest an alternative way by which the evolutionary relationship between humans and the apes can be determined by molecular techniques. [1]

- **Comparing DNA sequences of other proteins;; OR**
- **Compare mRNA sequences of other proteins;; OR**
- **Compare amino acid sequences of other proteins;;**

[Total: 10]

4. The figure below shows how a one-celled zygote undergoes various cellular divisions to give rise to other cell types.



(a) Describe two unique features of stem cells that could be seen from the figure above. [2]

- **Stem cells can undergo differentiation to give rise to specialized cell types;;**
- **Stem cells are capable of dividing indefinitely and producing copies of themselves / capable of self renewal;;**

(b) Describe the developmental potential of the one-celled zygote and the inner cell mass of the blastocyst. [4]

- **One celled zygote is a totipotent stem cell;;**
- **that can form extraembryonic membranes and the embryo;;**
- **Inner cell mass of the blastocyst are all pluripotent stem cells;;**
- **that can form all/most cell types of a human.;;**

(c) Multipotent stem cells can form specialized cell types of the tissue in which they reside. Give an example of a multipotent stem cell and the specialized cell types that it can form. [2]

- **Hematopoietic stem cells.;;**
 - **They can give rise to red blood cells and white blood cells.;;**
- OR**
- **Bone marrow stromal stem cells.;;**
 - **They can give rise to bone cells/osteocytes, cartilage cells/chondrocytes, fat cells/adipocytes and other kinds of connective tissue cells such as those in tendons.;;**
- OR**
- **Neural stem cells.;;**

- They can give rise to neurons and non-neuronal cells – astrocytes and oligodendrocytes.;;

(d) Adult stem cells are multipotent stem cells found among differentiated cells in a tissue or organ. What are the primary roles of these adult stem cells and describe one appropriate unique feature of this type of stem cells that is not mentioned in (a). [2]

- Their primary role is to maintain and repair the specific tissues in which they reside.;;
- They can remain undifferentiated / unspecialized for long periods of time.;;

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

You answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

5.

(a) Describe the structure of DNA and its arrangement in a chromosome. [8]

1. Double helix;;
 2. 2 anti - parallel strands with 3' and 5' ends;;
 3. Specific nucleotide sequence along each DNA strand;;
 4. Each nucleotide consists – deoxyribose sugar, phosphoric acid and nitrogenous base;;
 5. 4 bases - Adenine, Thymine, Cytosine and Guanine (spell out for the first time);;
 6. Nitrogenous bases projecting at right angle within the helix;;
 7. 0.34nm between adjacent base pairs;;
 8. Hydrogen bonds between base pairs;;
 9. Complementary base-pairing: A with T, C with G;;
 10. Sugar-phosphate backbone with phosphodiester links (bonds);;
 11. Width between the two sugar-phosphate backbone is 0.2nm;;
 12. 10 bases pairs in one complete turn of helix / 3.4nm per turn;;
- [Total marks = 12, max 4 marks]

13. DNA wound around histone octamer/histones to form nucleosome;;
 14. Formation of 'beads on a string';;
 15. Strings of nucleosomes coiled to form a 30nm chromatin fiber / solenoid;;
 16. Looped domains of the 30nm fiber / 30nm fiber forms loops;;
 17. Attached to a scaffold of non-histone proteins;;
 18. Further looping and coiling to form metaphase chromosome;;
- [Total marks = 6, max 4 marks]

[Total marks = 18, max 8 marks]

(b) Compare and contrast between DNA replication and transcription. [12]

Similarities:

1. Both involve unwinding of the double helix DNA::
2. Both involve breaking of the weak hydrogen bonds between complementary bases::
3. Both involve formation of phosphodiester bonds between adjacent nucleotides::
4. Both involve aligning free nucleotides through complementary base pairing::
5. Both products elongate in 5' to 3' direction::
6. Both occur in the nucleus::

[Total marks = 6, max 5 marks]

Differences:

Features	DNA replication	Transcription
1. (Main) Enzyme involved	<u>DNA polymerase</u> - binds to the parental DNA molecule at the origin of replication;	<u>RNA polymerase</u> - binds to the promoter, a region near the beginning of the gene to be transcribed;
2. Raw Material	Deoxyribonucleotides;	Ribonucleotides;
3. Template	Both strands of DNA molecule act as a template;	One strand of DNA molecule acts as the template;
4. Base Pairing	Adenine with Thymine and Cytosine with Guanine;	Adenine with Uracil and Cytosine with Guanine;
5. Presence of proofreading property on enzyme involved	DNA polymerase proofreads the newly synthesized daughter strand, ensuring precise complementary base pairing;	No proof-reading function;
6. Products	2 DNA molecules;	1 RNA molecule;
7. Location of the products	Products remain in the nucleus;	Products leave the nucleus;
8. Post – transcriptional modification	Products do not go through post - transcriptional modification;	Products / pre-mRNA may go through post-transcription modification / splicing / capping / poly-A tailing;
9. When the process occur	Prior to nuclear division / mitosis / meiosis;	Prior to translation;
10. Purpose	Double the amount of DNA so that after mitosis, the daughter cells will have the same amount of DNA as the parent cell / after meiosis, the daughter cells will have half the amount of DNA as the	For subsequent protein synthesis;

[Total marks = 10, max 7 marks]

If no point-by-point comparison for 'differences', to half total no. of marks obtained.

OR

6.
(a) Describe the polymerase chain reaction and explain the advantages and limitations of this procedure. [12]

1. in vitro technique to amplify specific sequence of DNA in a short period of time;

2. 3 step process (denaturation, hybridization / annealing, extension / elongation);

3. carried out in thermal cycler;

4. Denaturation;

5. heat reaction tube containing specific target dsDNA to ~ 94 –96°C;

6. to break H- bonds;

7. separating dsDNA into ssDNA;

8. Hybridisation / annealing;

9. cool reaction mixture to ~ 60 – 65°C;

10. allow DNA (forward and reverse) primers to complementary base-pair with ssDNA template;

11. Extension/elongation;

12. raise temperature to ~ 70 – 75°C;

13. Taq / DNA polymerase binds to DNA template where primer has annealed;

14. add complementary nucleotides to 3' OH ends of primers;

15. catalyses phosphodiester bonds between adjacent nucleotides;

16. process repeats itself to produce multiple copies of target sequence;

[Total marks = 8, max 6 marks]

ADVANTAGES (with elaboration):

1. Highly sensitive;;
- It allows the investigation of minute samples of DNA.
 - In criminal investigation, DNA fingerprints are prepared from the cells in a tiny speck of dried blood or at the base of a single human hair.
 - In pediatric medicine, physicians can detect genetic defects in very early embryos by collecting a few sloughed-off cells and amplifying their DNA.
 - PCR could also be used to examine the DNA of historical figures, such as Abraham Lincoln, or that of now-extinct species, as long as even a minuscule amount of their DNA remains intact.

2. Speed and ease of use;;

- DNA cloning by PCR can be performed in a few hours, using relatively unsophisticated equipment.
- A PCR reaction consists of 30 cycles with an individual cycle typically taking 3-5 min in an automated thermal cycler. This compares favorably with the time required for cell-based DNA cloning, which may take weeks.

3. Robustness;;

- PCR can permit amplification of specific sequences from material in which the DNA is badly degraded / embedded in a medium from which conventional DNA isolation is problematic.
- As a result, it is again very suitable for molecular anthropology and paleontology studies, for example the analysis of DNA recovered from archaeological remains. It has also been used successfully to amplify DNA from formalin-fixed tissue samples, which has important applications in molecular pathology and genetic linkage studies.

[Total marks = 3, max 3 marks]

LIMITATIONS (with elaboration):

1. Too sensitive;;

- Even a tiny amount of contaminant DNA in a sample may become amplified if it includes a DNA sequence complementary to the primers, potentially leading to an erroneous conclusion.

2. Need for target DNA sequence information;;

- In order to construct specific oligonucleotide primers that permit selective amplification of a particular DNA sequence, some prior sequence information is necessary.
- This normally means that the DNA region of interest has been partly characterized previously, often following cell-based DNA cloning.

3. Short size and limiting amounts of PCR product;;

- A clear disadvantage of PCR as a DNA cloning method has been the size range of the DNA sequences that can be cloned.
- Unlike cell-based DNA cloning where the size of cloned DNA sequences can approach 2 Mb, reported DNA sequences cloned by PCR have typically been in the 0.1-5 kb size range, often at the lower end of this scale.

4. Infidelity of DNA replication;;

- When DNA is replicated *in vitro* the copying error rate is considerably greater.
- Tag DNA polymerase has no associated 3'→5' exonuclease to confer a proofreading function, and the error rate due to base misincorporation during DNA replication is rather high.
 - ⇒ E.g. for a 1 kb sequence that has undergone 20 effective cycles of duplication, approximately 40% of the new DNA strands synthesized by PCR using this enzyme will contain an incorrect nucleotide resulting from a copying error.
- This means that, even if the PCR reaction involves amplification of a single DNA sequence, the final product will be a mixture of extremely similar, but not identical DNA sequences.

[Total marks = 4, max 3 marks]

(b) Discuss the goals and implications of the human genome project, including the benefits and ethical concerns. [8]

Goals

- To obtain a genetic linkage map of human genome.;;
This involved the identification of thousands of genetic markers and their localization along the autosomes and sex chromosomes.
- To obtain the DNA sequence of the entire human genome.;;
The entire genome is approximately 3.2 billion nucleotide base pairs in length.
- To develop technology for the management of human genome information.;;
The Human Genome Project has developed user-friendly tools to provide scientists easy access to up-to-date information obtained from the project. The Human Genome Project also has developed analytical tools to interpret genome information.
- To analyse the genomes of other model organisms.;;
These include bacterial species (e.g. *Escherichia coli* and *Bacillus subtilis*), *Drosophila melanogaster* (fruit fly), *Caenorhabditis elegans* (a nematode), *Arabidopsis thaliana* (a simple plant) and *Mus musculus* (mouse).
- To develop technological advances in genetic methodologies.;;
Some of the efforts of the Human Genome Project have involved improvements in molecular genetic technology such as gene cloning, DNA sequencing and so forth. The project has also been aimed at developing computer technology for data processing, storage and the analysis of sequence information.

[Total marks = 5, max 4 marks]

Benefits (with elaboration):

- The detection of mutant genes, which causes disease, could make the diagnosis of certain diseases easier and could make it possible accurately to predict the onset of diseases.;;
- Provide insight into the types of protein encoded by these genes. Thus increase the potential to treat genetic and acquired diseases by gene therapy/whereby defective genes are either replaced by normal genes or supplemented with other genes to prevent a disease taking its course.;;

Ethical issues (with elaboration):

- ⇒ Ownership and use of personal genetic information.;;
Who should have access to it, are a matter for concern e.g. Should employers/insurance agents have access to the genetic fingerprint and hire workers/granting insurance policies based on their genetic predisposition.
- ⇒ Reproductive issues regarding use of genetic information in reproductive decision making and reproductive rights.;;

There are concerns that prenatal genetic testing could lead to genetic manipulation or a decision to abort based on undesirable traits disclosed by the tests.

⇒ Clinical issues.;;

There are questions on the accuracy, reliability and utility of the genetic tests, how prepare are the healthcare professionals for the new genetics and how much can the interpretation be trusted.

⇒ Treatment versus Enhancement of humans.;;

Treatment includes administering to the disease whereas enhancement means to change the human genome to make it flawless. Hence instead of using HGP knowledge in treatment, people may use it for human enhancement instead and this will open up even more ethical issues for debate.

[Total marks = 6, max 4 marks]